ZOO404 PHYSIOLOGY

FINAL TERM PAPER PREPARATION BEST FILE BY

SULMAN ALI

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What are the conformers and its types

Animals that conform to external changes and Are unable to maintain homeostasis for internal conditions.

Types: 1= Osmoconformers 2= Oxyconformers 3= Thermoconformers

What is the purpose of ultrafiltrate or ultera secretion:

Ultrafiltration (UF) is used to remove essentially all colloidal particles (0.01 to 1.0 microns) from water and some of the largest dissolved contaminants. The pore size in a UF membrane is mainly responsible for determining the type and size of contaminants removed.

How does the Resistance to Flow is Overcome by Pressure:

When heart contracts, pressure in the ventricles increases.This pressure is used to overcome resistance to flow through the vessels.

Role of ADH:

Amount of water reabsorbed and conc. of urine is controlled by antidiuretic hormone. ADH acts by increasing water permeability of renal tubules.

How B12 is absorbed In small Intestine?

Vitamin B12 (molecular weight 1357) is the largest water-soluble essential nutrient taken up intact. Absorbed across the intestinal lumen in distal ileum by a mechanism independent of intrinsic factor.

Define donnan effect?

Donnan equilibrium implies that: if there is a nondiffusible solute in one of the sol. separated by semi-permeable membrane, the conc. of sol. on both sides do not equalize. Conc. of sol. with non-diffusible solute remains high even at equilibrium. This is Donnan effect.

Minns ornithine area cycle?

Ornithine-Urea Cycle is used by all ‘'vertebrates except teleost fishes. it Synthesize urea primarily in the liver. Two ammonia and one CO2 molecule are added to ornithine to form Arginine. Enzyme arginase cleaves arginine to remove a molecule of urea and regenerate ornithine.

Differentiate between Endopeptidases and Exopeptidases:

Endopeptidases	Exopeptidases
Endopeptidases Attack peptide bonds within protein molecule.	Exopeptidases Attack terminal peptide bonds of shorter polypeptides.
Break large peptide chains into shorter polypeptide segments	Release free amino acids or dipeptides and tripeptides

Write the factors of ultra filtration?

Filtration takes place entirely passively across three layered wall of the glomerulus into the capsular space.

Filtration depends on three factors: 1= Net hydrostatic pressure 2= Colloidal osmotic pressure 3= Permeability of filtration membrane.

Write a note on lipases:

Lipases are Lipid and fat digesting enzymes. Lipases Secreted in pancreatic juice and by intestinal epithelium and Act on emulsified fat droplets and hydrolyze them into: • Monoglycerides • Diglycerides • Free fatty acids and glycerol.

Emulsification by bile salts is necessary for lipases to hydrolyze lipids • If bile salts are insufficient, fats are not emulsified • Fat digestion by lipases remains incomplete. These are enzymes that the body uses to break down fats in food so they can be absorbed in the intestines. Lipase is produced in the pancreas, mouth, and stomach. For example Candida rugosa.

Write a Note on blood vessels?

Blood vessels carry blood during circulation.

Three types: 1= Arteries 2= Veins 3= Capillaries

1= Arteries: Arteries carry blood away from heart to organs and tissues. Or Arteries, part of your circulatory (cardiovascular) system, are the blood vessels that bring oxygen-rich blood from your heart to all of your body's cells.

2= Veins: Veins return blood to heart. Or Veins are a type of blood vessel that return deoxygenated blood from your organs back to your heart.

3= Capillaries: Capillaries have thin walls formed of only a single layer of endothelial cells. Capillaries permit exchange of materials between blood and tissues. Or Capillaries are delicate blood vessels that exist throughout your body. They transport blood, nutrients and oxygen to cells in your organs and body systems. Capillaries are the smallest blood vessels in your vascular system.

How the kidney regulate the PH:

Kidneys maintain pH by two ways: 1= Reabsorption of bicarbonate ions 2= Removal of fixed acids and H+

Role of ph in body:

PH is the measure of how acidic or basic a substance is. In humans, pH balance plays a role in keeping the body functioning optimally. The ideal pH of the body is slightly alkaline, which facilitates certain biochemical reactions like oxygenating the blood.

Effect of Catecholamines on Pacemaker cells:

Adrenergic nerve fibers innervate S-A node, atria, A-V node and ventricles. Norepinephrine is released from them. Norepinephrine increases the rate of contraction of myocardium. It increases Na+ and Ca2+ conductance. Na+ and Ca2+ accelerate pacemaker depolarization.

Reabsorption of water occurs in which part of the loop of henle:

Reabsorption of water continues in descending limb. It is freely permeable to water.

Reabsorption at DCT:

Only 15–20 % of initial filtrate volume reaches DCT. Tubular cells of DCT, actively reabsorb Na+ and Cl-. Distal portions of DCT also contain ion pumps that reabsorb tubular Na+ in exchange for K+. DCT also reabsorbs HCO3 - ions to regulate urine pH.

Function of acetylcholine:

ACh increases chromaffin cells’ conductance for Ca2+.

What is the Alimentary Canal?

The alimentary canal is mainly referred to as the pathway by which food enters our body and moves out through the anus after digestion. It is a tube-like structure which starts from the mouth and ends in the anus. The alimentary canal plays a primary role in human digestion and is also termed as the digestive tract.

Organs of the Alimentary Canal

The main organs of the alimentary canal are: 1= The Mouth and Oral cavity. 2= Oesophagus. 3= Stomach. 4= Small intestine. 5= Large intestine. 6= anus (note ye long he is liye thora thora explain kr daina ap sb in sbhi parts ko apny pas sy)

Write two mechanisms of motility?

Alimentary canal motility is achieved by two different mechanisms:

1= Muscular motility: Involves contractions of muscles in the walls of alimentary canal. It Permits handling of harder and larger pieces of food. 2= Ciliary motility: Involves beating of cilia Lining the digestive tract. Cilia generate fluid.

Types of motality: There are different types of motility, often classified as swimming, swarming, twitching, gliding, and sliding

What are the Hormones affecting kidney function and which hormone produce kidney:

Hormones affecting kidney function including: Antidiuretic hormone (ADH) • Renin-AngiotensinAldosterone System • Atrial natriuretic peptide (ANP).

The kidneys make two main hormones, vitamin D and erythropoietin.

Steroid Hormones:

Steroid hormones are synthesized from cholesterol. They consist of four fused carbon rings including three cyclohexyl rings and one cyclopentyl ring combined into a single structure.

Cisternae of Sarcoplasmic Reticulum:

Terminal cisternae are larger chambers that are closely associated with Ttubules (12 nm apart). These are the primary site of calcium release as they have channel proteins that open with AP in T tubules.

Digestive mechanism of stomach:

The stomach is a hollow organ, or "container," that holds food while it is being mixed with stomach enzymes. These enzymes continue the process of breaking down food into a usable form. Cells in the lining of your stomach secrete a strong acid and powerful enzymes that are responsible for the breakdown process. When the contents of the stomach are processed enough, they’re released into the small intestine.

How the impuls generate from heart:

An electrical stimulus is generated by the sinus node (also called the sinoatrial node, or SA node). This is a small mass of specialized tissue located in the right upper chamber (atria) of the heart. The sinus node generates an electrical stimulus regularly, 60 to 100 times per minute under normal conditions.

What is glomerular filtration?

Glomerular filtration is the first step in making urine. It is the process that your kidneys use to filter excess fluid and waste products out of the blood into the urine collecting tubules of the kidney, so they may be eliminated from your body.

Describe the absorptive Surface am in small intestine?

Epithelial lining of intestine is specialized for absorptive functions. It has:

1= Increased surface area for absorption

2= Rich supply of transport system

The adaptations to increase Surface area art enhanced by anatomical features:

Intestinal folds, Villi, Epithelial Cells of Villi, Microvilli

Intestinal folds, Villi and microvilli collectively increase the surface area more than 500 times. The Total area for absorption in small intestine of human is 200-300 m2

Comparison: gross cylindrical area of small intestine is only 0.4 m2.

Name the constitute of the lungs:

Terminal bronchioles • Respiratory bronchioles • Alveolar ducts • Alveolar sacs

Define hypercapnia:

A condition of abnormally elevated carbon dioxide levels in the blood is called hypercapnia. Hypercapnia results in an increase in ventilation. Response occurs almost immediately.

how neurotransmitters alter the permeability of synaptic region:

Neurotransmitters induce large increase in permeability to ions by acting on integral membrane proteins in nerve and muscle cells.

Quantal Release of Neurotransmitter:

Neurotransmitters are generally released in tiny packets called quanta. Each quantum may consist of about 2000 to 10,000 molecules of transmitter molecules.

Significance of Motility:

It is significant for: • Translocation and conduction of food • Mechanical treatment of food • Prevent scumming.

P wave and p-r interval:

P wave	p-r interval
Caused by electrical potentials when atria depolarize	Period of time from start of P wave to beginning of the QRS complex.
Occurs just prior to contraction of atria	Indicates amount of time required for SA depolarization to reach the ventricles
Ventricles are in diastole during the expression of P wave.	

What is a Electrocardiogram:

An electrocardiogram (ECG) is a simple test that can be used to check your heart's rhythm and electrical activity. Sensors attached to the skin are used to detect the electrical signals produced by your heart each time it beats.

Write a not on appendix:

The appendix is a narrow, finger-shaped pouch that projects out from the colon. Appendicitis occurs when the appendix becomes inflamed and filled with pus. Appendicitis is an inflammation of the appendix, a finger-shaped pouch that projects from your colon on the lower right side of your abdomen.

Uses of Kymograph:

Study of skeletal muscle contractions (twitch and tetanus) • Cardiac muscle activities (cardiac cycle) • Measurement of blood pressure • Rate of respiration

Peristalsis and reverse Peristalsis:

Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract. It starts in the esophagus where strong wave-like motions of the smooth muscle move balls of swallowed food to the stomach.

Retroperistalsis is the reverse of the involuntary smooth muscle contractions of peristalsis. It usually occurs as a precursor to vomiting. Local irritation of the stomach, such as bacteria or

food poisoning, activates the emetic center of the brain which in turn signals an imminent vomiting reflex.

function of triad: The triad is a skeletal muscle substructure responsible for the regulation of excitation-contraction coupling. It is formed by the close apposition of the T-tubule and the terminal sarcoplasmic reticulum.

Write a note on pH regulation?

PH regulation is The processes operating in living organisms to regulate the concentration of hydrogen ions and preserve a viable acid-base state.

Regulation of body fluid pH is one of the most important physiological functions at homeostasis, because activity of most chemical reactions via enzyme protems is dependent on fluid PH

PH of blood is maintained within a narrow alkaline range 7.35 to 7_45. Outside this range. pH becomes incompatible with life. Enzymes and proteins are denatured.

Kidneys maintain PH by two ways:

- Reabsorption of bicarbonate ions
- Remo% al of fixed acids and 11